K.R. Mangalam University

School of Engineering & Technology

Operating System Lab (ENCA252)

Lab Assignment 1

Implementation and Analysis of CPU Scheduling Algorithms (FCFS & SJF)

Submitted by:

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Submitted To:

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1. Objective

To implement and analyze CPU Scheduling Algorithms — **First Come First Serve (FCFS)** and **Shortest Job First (SJF)** — and evaluate their performance using standard metrics such as Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT).

2. Tools & Technologies Used

Tool	Purpose
Python 3.x	Implementation language
Built-in libraries: `os`, `copy`, `time`	Core utilities
Linux / macOS Terminal	Execution environment
GitHub	Version control and submission

3. Key Concepts

Term	Formula	Description
CT	—	Time when process finishes execution
TAT	CT − AT	Total time from arrival to completion
WT	TAT − BT	Time the process spent waiting in the queue

4. Task 1 — Process Creation & Input Handling

Each process is modelled using a Python class with three attributes:

- **PID** — Process Identifier
- **AT** — Arrival Time
- **BT** — Burst Time

Processes are collected from the user and stored in a list, then displayed in tabular format.

Input Table

PID	Arrival Time (AT)	Burst Time (BT)
P1	0	5
P2	1	3
P3	2	8
P4	3	2

Screenshot — Input Table

```
bhavya@bhavya-Latitude-7390-2-in-1:~/Documents$ cd /home/bhavya/Documents/os/os_assignment_1
bhavya@bhavya-Latitude-7390-2-in-1:~/Documents/os/os_assignment_1$ nano Lab_Assignment_1.py
bhavya@bhavya-Latitude-7390-2-in-1:~/Documents/os/os_assignment_1$ python3 Lab_Assignment_1.py

=====================================================
       CPU SCHEDULING SIMULATOR
=====================================================

Enter number of processes (min 4): 4

Enter Arrival Time and Burst Time for each process:
---------------------------------------------
  P1 → Arrival Time: 0
  P1 → Burst Time:   5

  P2 → Arrival Time: 1
  P2 → Burst Time:   3

  P3 → Arrival Time: 2
  P3 → Burst Time:   8

  P4 → Arrival Time: 3
  P4 → Burst Time:   2


=====================================================
  INPUT: PROCESS TABLE
=====================================================
  PID    Arrival Time    Burst Time
---------------------------------------
  P1     0               5
  P2     1               3
  P3     2               8
  P4     3               2
---------------------------------------
```

5. Task 2 — FCFS Scheduling

First Come First Serve (FCFS) is the simplest non-preemptive scheduling algorithm. Processes are executed in the order of their arrival time. If the CPU is idle (no process has arrived), it waits until the next process arrives.

FCFS Results Table

PID	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	1	3	8	7	4
P3	2	8	16	14	6
P4	3	2	18	15	13

Screenshot — FCFS Output

```
=========================================================
  FCFS — Results
=========================================================
  PID    AT     BT     CT     TAT    WT
---------------------------------------------
  P1     0      5      5      5      0
  P2     1      3      8      7      4
  P3     2      8      16     14     6
  P4     3      2      18     15     13
---------------------------------------------
```

6. Task 3 — SJF Scheduling (Non-Preemptive)

Shortest Job First (SJF) selects the process with the smallest burst time from all currently arrived processes. It is non-preemptive — once a process starts, it runs to completion.

SJF Results Table

PID	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P4	3	2	7	4	2
P2	1	3	10	9	6
P3	2	8	18	16	8

Screenshot — SJF Output

```
=========================================================
  SJF — Results
=========================================================
  PID    AT     BT     CT     TAT    WT
---------------------------------------------
  P1     0      5      5      5      0
  P4     3      2      7      4      2
  P2     1      3      10     9      6
  P3     2      8      18     16     8
---------------------------------------------
```

7. Task 4 — Gantt Chart Representation

FCFS Gantt Chart

```
| P1      | P2      | P3      | P4      |
0         5         8        16        18
```

SJF Gantt Chart

```
| P1      | P4      | P2      | P3      |
0         5         7        10        18
```

Screenshot — Gantt Charts

```
============================================================
  TASK 4 — GANTT CHARTS
============================================================

  FCFS Gantt Chart
  --------------------------------
  |   P1   |   P2   |   P3   |   P4   |
  --------------------------------
  0       5       8       16      18

  SJF Gantt Chart
  --------------------------------
  |   P1   |   P4   |   P2   |   P3   |
  --------------------------------
  0       5       7       10      18
```

8. Task 5 — Performance Analysis & Comparison

Metric	FCFS	SJF
Average Turnaround Time (TAT)	10.25	8.50
Average Waiting Time (WT)	5.75	4.00

Analysis

- **SJF** produces a **lower average waiting time** compared to FCFS.
- FCFS can cause the **Convoy Effect**, where short processes get stuck waiting behind long ones.
- SJF is more efficient but requires **prior knowledge of burst times**, which is not always available in real systems.

Conclusion

SJF is the better algorithm for minimizing average waiting time and turnaround time. However, FCFS is simpler to implement and fairer in terms of arrival order. The choice depends on the use case and available information about process burst times.

Screenshot — Performance Comparison

```
============================================================
  PERFORMANCE COMPARISON: FCFS vs SJF
============================================================
  Metric                          FCFS       SJF
------------------------------------------------------
  Average Turnaround Time (TAT)   10.25      8.50
  Average Waiting Time (WT)       5.75       4.00
------------------------------------------------------

  📊 ANALYSIS:
  ✅ SJF has LOWER average waiting time.
     → SJF is better for minimizing wait time.
     → It reduces idle CPU time by running short jobs first.

  ℹ  TRADE-OFFS:
  • FCFS is simple but can cause the 'Convoy Effect'
    (long jobs blocking short ones).
  • SJF minimizes average waiting time but requires
    knowing burst times in advance (not always practical).
============================================================

  ✅ Simulation Complete!

bhavya@bhavya-Latitude-7390-2-in-1:~/Documents/os/os_assignment_1$
```

9. GitHub Repository

~~link-~~ github

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10. Source Code

python

```python
# Full source code available in the GitHub repository linked above.
# File: cpu_scheduling.py

# ============================================================
#  OS Lab Assignment 1 — CPU Scheduling: FCFS & SJF
#  Course: ENCA252 | Program: BCA (AI & DS)
# ============================================================

# ============================================================
# TASK 1: Process Class & Input Handling
# ============================================================

# A class is like a blueprint. Here we define what a "Process" looks like.
class Process:
    def __init__(self, pid, at, bt):
        """
        pid = Process ID       (e.g., P1, P2 ...)
        at  = Arrival Time     (when the process arrives in the ready queue)
        bt  = Burst Time       (how long it needs on the CPU)
        """
        self.pid = pid
        self.at  = at
        self.bt  = bt

        # These will be calculated later by the scheduling algorithms
        self.ct  = 0   # Completion Time
        self.tat = 0   # Turnaround Time = CT - AT
        self.wt  = 0   # Waiting Time    = TAT - BT


def get_processes():
    """Ask the user to enter process details and return a list of Process
objects."""

    print("\n" + "="*55)
    print("       CPU SCHEDULING SIMULATOR")
```

```
    print("="*55)

    # Ask how many processes the user wants to enter (must be at least 4)
    while True:
        try:
            n = int(input("\nEnter number of processes (min 4): "))
            if n < 4:
                print("  ⚠  Please enter at least 4 processes.")
            else:
                break
        except ValueError:
            print("  ⚠  Invalid input. Please enter a number.")

    processes = []  # Empty list; we'll fill it with Process objects

    print(f"\nEnter Arrival Time and Burst Time for each process:")
    print("-" * 45)

    for i in range(1, n + 1):
        while True:
            try:
                at = int(input(f"  P{i} → Arrival Time: "))
                bt = int(input(f"  P{i} → Burst Time:   "))
                if at < 0 or bt <= 0:
                    print("  ⚠  Times must be non-negative (Burst Time >
0).")
                    continue
                processes.append(Process(f"P{i}", at, bt))
                print()
                break
            except ValueError:
                print("  ⚠  Invalid input. Enter whole numbers only.")

    return processes


def print_table(processes, title="Scheduling Results"):
    """Print any list of processes in a neat formatted table."""

    print("\n" + "="*55)
    print(f"  {title}")
    print("="*55)
    print(f"  {'PID':<6} {'AT':<6} {'BT':<6} {'CT':<6} {'TAT':<6} {'WT':
<6}")
    print("-" * 45)
```

```
    for p in processes:
        print(f"  {p.pid:<6} {p.at:<6} {p.bt:<6} {p.ct:<6} {p.tat:<6} {p.wt:
<6}")

    print("-" * 45)


# ============================================================
# TASK 2: FCFS Scheduling (First Come First Serve)
# ============================================================

def fcfs(processes):
    """
    FCFS: The process that arrives FIRST gets the CPU first.
    It is NON-PREEMPTIVE — once a process starts, it runs to completion.

    Steps:
      1. Sort processes by Arrival Time (AT)
      2. Track current time; handle idle gaps
      3. Calculate CT, TAT, WT for each process
    """

    # We sort by arrival time — whoever comes first, runs first
    # We use 'key=lambda p: p.at' which means "sort by the .at attribute"
    sorted_procs = sorted(processes, key=lambda p: p.at)

    current_time = 0        # The "clock" of the CPU
    gantt = []              # Will store (process_name, start_time,
end_time)

    for p in sorted_procs:

        # If CPU is idle (current process hasn't arrived yet), fast-forward
time
        if current_time < p.at:
            gantt.append(("IDLE", current_time, p.at))  # Record idle period
            current_time = p.at

        start_time = current_time           # Process starts running now

        current_time += p.bt               # Process runs for its burst time

        p.ct  = current_time               # It finishes at current_time
        p.tat = p.ct - p.at                # Total time from arrival to
finish
        p.wt  = p.tat - p.bt               # Time it spent waiting
```

```
        gantt.append((p.pid, start_time, p.ct))

    return sorted_procs, gantt


# ============================================================
# TASK 3: SJF Scheduling (Shortest Job First — Non-Preemptive)
# ============================================================

def sjf(processes):
    """
    SJF: Among all ARRIVED processes, pick the one with the SMALLEST burst
time.
    It is NON-PREEMPTIVE — once chosen, it runs to completion.

    Steps:
      1. At each step, look at all processes that have arrived so far
      2. From those, pick the one with minimum Burst Time
      3. Run it fully, then repeat
    """

    # We work with copies so we don't mess up the original list
    import copy
    remaining = copy.deepcopy(processes)

    # Reset any previous scheduling results on original objects
    for p in processes:
        p.ct = p.tat = p.wt = 0

    current_time = 0
    completed    = []      # Processes that have finished
    gantt        = []      # Gantt chart data

    while remaining:

        # Find all processes that have ALREADY arrived by current_time
        available = [p for p in remaining if p.at <= current_time]

        if not available:
            # No process has arrived yet — CPU is idle
            # Jump time to the nearest arrival
            next_arrival = min(remaining, key=lambda p: p.at).at
            gantt.append(("IDLE", current_time, next_arrival))
            current_time = next_arrival
            continue
```

```
        # From available processes, pick the one with SHORTEST burst time
        # If two have equal BT, pick the one that arrived first (stable
sort)
        shortest = min(available, key=lambda p: (p.bt, p.at))

        start_time   = current_time
        current_time += shortest.bt

        shortest.ct  = current_time
        shortest.tat = shortest.ct - shortest.at
        shortest.wt  = shortest.tat - shortest.bt

        gantt.append((shortest.pid, start_time, shortest.ct))

        completed.append(shortest)
        remaining.remove(shortest)   # This process is done; remove from
queue

    # Now copy results back to the original process objects (matched by PID)
    for orig in processes:
        for done in completed:
            if orig.pid == done.pid:
                orig.ct  = done.ct
                orig.tat = done.tat
                orig.wt  = done.wt

    return completed, gantt


# ============================================================
# TASK 4: Gantt Chart (Text-Based Visualization)
# ============================================================

def print_gantt_chart(gantt, title="Gantt Chart"):
    """
    Draw a simple text-based Gantt chart showing execution order.

    Example output:
    | P1 | P3 | IDLE | P2 |
    0    3    7     10   14
    """

    print(f"\n  {title}")
    print("  " + "-" * (len(gantt) * 8))
```

```
    # Top row: process names in boxes
    row = "  |"
    for (name, start, end) in gantt:
        label = name.center(6)   # Center the name in a 6-char wide cell
        row += f" {label} |"
    print(row)

    print("  " + "-" * (len(gantt) * 8))

    # Bottom row: time markers
    time_row = "  "
    for (name, start, end) in gantt:
        time_row += str(start).ljust(8)   # Left-justify start times
    # Add the very last end time
    time_row += str(gantt[-1][2])
    print(time_row)


# ============================================================
# TASK 5: Performance Analysis & Comparison
# ============================================================

def performance_analysis(fcfs_procs, sjf_procs):
    """
    Calculate and compare average WT and TAT for both algorithms.
    Then give a conclusion on which is better and why.
    """

    n = len(fcfs_procs)

    # Calculate averages for FCFS
    fcfs_avg_tat = sum(p.tat for p in fcfs_procs) / n
    fcfs_avg_wt  = sum(p.wt  for p in fcfs_procs) / n

    # Calculate averages for SJF
    sjf_avg_tat  = sum(p.tat for p in sjf_procs) / n
    sjf_avg_wt   = sum(p.wt  for p in sjf_procs) / n

    print("\n" + "="*55)
    print("  PERFORMANCE COMPARISON: FCFS vs SJF")
    print("="*55)
    print(f"  {'Metric':<30} {'FCFS':<10} {'SJF':<10}")
    print("-" * 50)
    print(f"  {'Average Turnaround Time (TAT)':<30} {fcfs_avg_tat:<10.2f}
{sjf_avg_tat:<10.2f}")
    print(f"  {'Average Waiting Time (WT)':<30} {fcfs_avg_wt:<10.2f}
```

```
{sjf_avg_wt:<10.2f}")
    print("-" * 50)

    # Conclusion
    print("\n  📊 ANALYSIS:")
    if sjf_avg_wt < fcfs_avg_wt:
        print("  ✅ SJF has LOWER average waiting time.")
        print("     → SJF is better for minimizing wait time.")
        print("     → It reduces idle CPU time by running short jobs
first.")
    elif fcfs_avg_wt < sjf_avg_wt:
        print("  ✅ FCFS has LOWER average waiting time in this case.")
        print("     → This can happen when short jobs arrive late.")
    else:
        print("  Both algorithms performed equally in this case.")

    print("\n  ℹ  TRADE-OFFS:")
    print("  • FCFS is simple but can cause the 'Convoy Effect'")
    print("    (long jobs blocking short ones).")
    print("  • SJF minimizes average waiting time but requires")
    print("    knowing burst times in advance (not always practical).")
    print("="*55)


# ============================================================
# MAIN — Runs everything
# ============================================================

def main():

    # ---------- Task 1: Get process input ----------
    processes = get_processes()

    # Show the input table
    print("\n" + "="*55)
    print("  INPUT: PROCESS TABLE")
    print("="*55)
    print(f"  {'PID':<6} {'Arrival Time':<15} {'Burst Time':<12}")
    print("-" * 40)
    for p in processes:
        print(f"  {p.pid:<6} {p.at:<15} {p.bt:<12}")
    print("-" * 40)

    # ---------- Task 2: Run FCFS ----------
    import copy
    fcfs_procs_input = copy.deepcopy(processes)   # Fresh copy for FCFS
```

```python
    sjf_procs_input  = copy.deepcopy(processes)   # Fresh copy for SJF

    fcfs_result, fcfs_gantt = fcfs(fcfs_procs_input)
    print_table(fcfs_result, title="FCFS — Results")

    # ---------- Task 3: Run SJF ----------
    sjf_result, sjf_gantt = sjf(sjf_procs_input)
    print_table(sjf_result, title="SJF — Results")

    # ---------- Task 4: Gantt Charts ----------
    print("\n" + "="*55)
    print("  TASK 4 — GANTT CHARTS")
    print("="*55)
    print_gantt_chart(fcfs_gantt, title="FCFS Gantt Chart")
    print_gantt_chart(sjf_gantt,  title="SJF Gantt Chart")

    # ---------- Task 5: Performance Analysis ----------
    performance_analysis(fcfs_result, sjf_result)

    print("\n  ✅ Simulation Complete!\n")


# This line means: only run main() if THIS file is executed directly
# (not if it's imported by another script)
if __name__ == "__main__":
    main()
```